

Vulnerability Assessment Report

Elevate Labs — Cybersecurity Internship | Task 3

Target	localhost (127.0.0.1)	Date	June 3, 2026
System	Ubuntu 24.04 — Aspire AS15-42	Assessor	Naveen V
Tool	Nmap 7.94SVN --script vuln	Status	Completed

Executive Summary

A vulnerability scan was performed on a personal Ubuntu machine using Nmap with NSE vulnerability scripts. Two services were discovered running without user awareness — CUPS (printer service) on port 631 and MySQL on port 3306. Three vulnerabilities were identified on CUPS: one rated High (CVSS 7.5), one confirmed exploitable HTTP Verb Tampering, and one Medium-severity Slowloris DoS. All vulnerabilities were successfully remediated using defense-in-depth. A final rescan confirmed zero open ports — fully hardened system.

Findings Summary

#	Vulnerability	Severity	CVSS	Service	Status
1	CVE-2024-47850	High	7.5	CUPS 2.4 / port 631	Fixed
2	HTTP Verb Tampering	High	N/A	CUPS 2.4 / port 631	Fixed
3	Slowloris DoS (CVE-2007-6750)	Medium	5.0	CUPS 2.4 / port 631	Fixed
4	MySQL Info (NODEJS:602)	Low	0.0	MySQL 8.0.46 / port 3306	Fixed

Detailed Findings

Vulnerability 1 — CVE-2024-47850

CVE ID	CVE-2024-47850
CVSS Score	7.5 — High
Service	CUPS 2.4 on port 631
Description	Critical vulnerability in CUPS printer service allowing remote exploitation by unauthenticated attackers.

Impact	Remote attacker can exploit the service without authentication.
Fix Applied	CUPS service stopped, disabled (systemctl disable), and port 631 blocked via UFW firewall.

Vulnerability 2 — HTTP Verb Tampering

Reference	CAPEC-274 / OWASP
State	VULNERABLE — Exploitable (Confirmed)
Service	CUPS 2.4 on port 631 /admin
Description	Web server only restricts GET/POST methods. Attacker uses PUT, DELETE, or HEAD requests to bypass authentication on the /admin panel.
Impact	Attacker can access admin panel without any credentials.
Fix Applied	CUPS service completely disabled — attack surface fully eliminated.

Vulnerability 3 — Slowloris DoS (CVE-2007-6750)

CVE ID	CVE-2007-6750
Severity	Medium — Likely Vulnerable
Service	CUPS 2.4 on port 631
Description	Attacker opens many partial HTTP connections to exhaust server resources, causing Denial of Service.
Impact	CUPS service can be crashed or made unavailable.
Fix Applied	CUPS service disabled — no running service means no DoS target.

Vulnerability 4 — MySQL Informational (NODEJS:602)

ID	NODEJS:602
CVSS Score	0.0 — Informational only
Service	MySQL 8.0.46 on port 3306
Description	Informational finding only. No exploitable vulnerability confirmed on this MySQL version.
Impact	None — no real exploit available.
Fix Applied	MySQL service stopped and disabled as security best practice.

Remediation Applied

Defense-in-depth approach — 3 layers of protection applied:

Layer	Action	Command
Layer 1	Stop services immediately	<code>sudo systemctl stop cups / mysql</code>
Layer 2	Disable auto-start on boot	<code>sudo systemctl disable cups / mysql</code>
Layer 3	Block ports via firewall	<code>sudo ufw enable && sudo ufw deny 631</code>

Before vs After

Metric	Before Scan	After Remediation
Open Ports	2 (631, 3306)	0
High Vulnerabilities	2	0
Medium Vulnerabilities	1	0
Low / Informational	1	0
Attack Surface	CUPS + MySQL exposed	Fully closed

Key Learning

Services installed for past projects (such as MySQL) continue running in the background after project completion — a very common real-world security risk. CUPS was also running for over 2.5 hours without user awareness, exposing three vulnerabilities. Regular vulnerability scanning helps identify and eliminate forgotten attack surfaces before attackers can exploit them.

Concept	Definition
Vulnerability Scanning	Automated detection of known security weaknesses in a system
CVE	Common Vulnerabilities and Exposures — unique ID assigned to each known vulnerability
CVSS	Common Vulnerability Scoring System — rates severity from 0.0 (none) to 10.0 (critical)
Attack Surface	All possible entry points an attacker could use to access a system
Defense in Depth	Applying multiple independent layers of security controls
False Positive	Scanner reports a vulnerability that does not actually exist on the system
Remediation	The process of fixing or mitigating an identified vulnerability